

Name:.....

Student
Number

Teacher:.....



Pymble Ladies' College

Mathematics Extension 1

HSC Trial Examination

Term 3 2025

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks 70

SECTION 1 – 10 marks (pages 1-5)

- Attempt Questions 1-10
- Allow about 15 minutes for this section
- Answer each question on the multiple-choice answer sheet provided in the answer booklet.

SECTION II – 60 marks (pages 7-14)

- Attempt Questions 11-14
- Allow about 1 hours and 45 minutes for this section
- Answer each question in the appropriate space in the Answer Booklet. Extra writing pages are included at the end of each question.

SECTION I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1. What are the values of x for which $(x+3)(x-1)(2-x) > 0$?

(A) $x < -3$ or $1 < x < 2$
(B) $x > -3$ or $1 < x < 2$
(C) $-3 < x < 1$ or $x > 2$
(D) $-3 < x < 1$

2. If $x+a$ is a factor of $4x^5 - 11ax^3 + x^4$, $a \neq 0$, what is the value of a ?

(A) -2
(B) 3
(C) 2
(D) -3

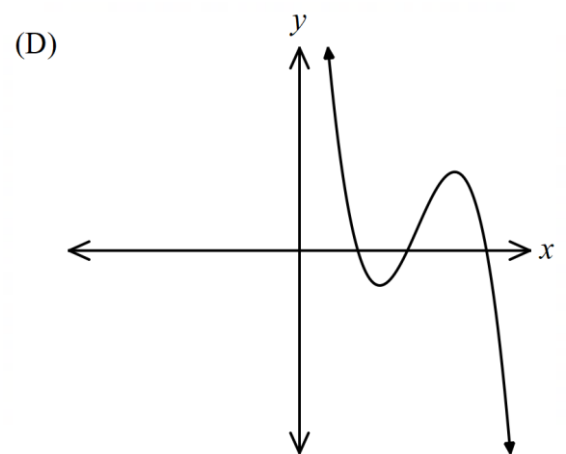
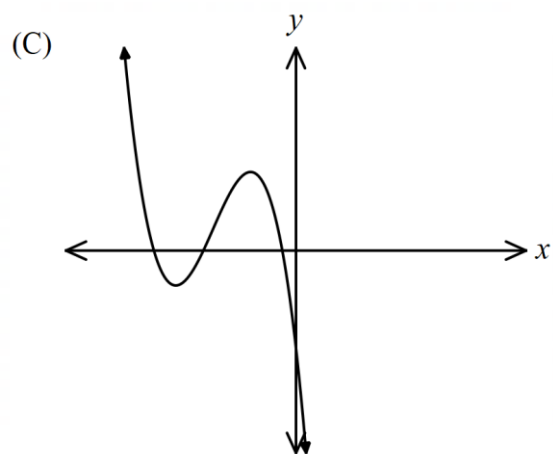
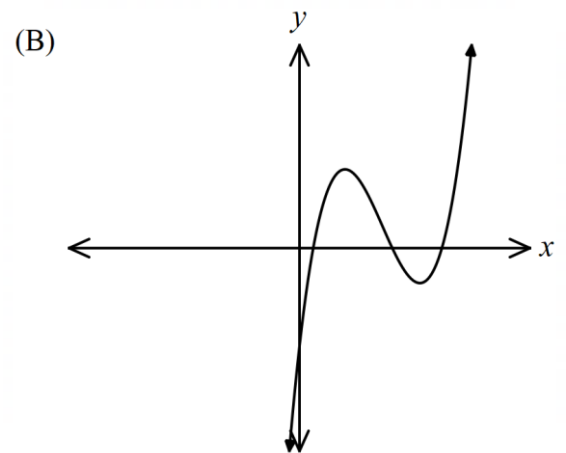
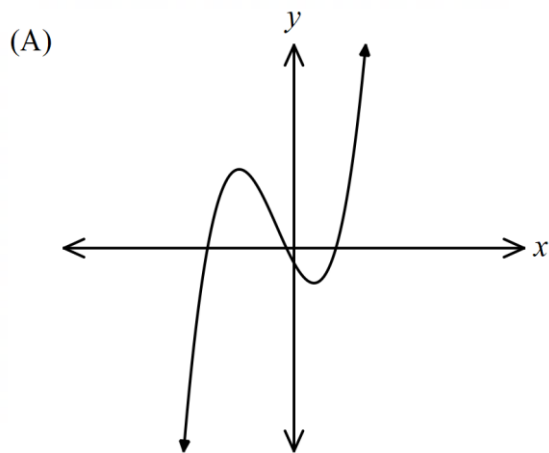
3. What is the area of the region enclosed by the graphs of $y = 4 - x^2$ and $y = 3$?

(A) $\frac{2}{3}$ square units
(B) $\frac{4}{3}$ square units
(C) $\frac{7}{3}$ square units
(D) $\frac{8}{3}$ square units

4. $\overrightarrow{AB} = \tilde{i} + 3\tilde{j}$ and $\overrightarrow{BC} = 2\tilde{i} - 2\tilde{j}$. What is the magnitude of $\overrightarrow{AC}$?

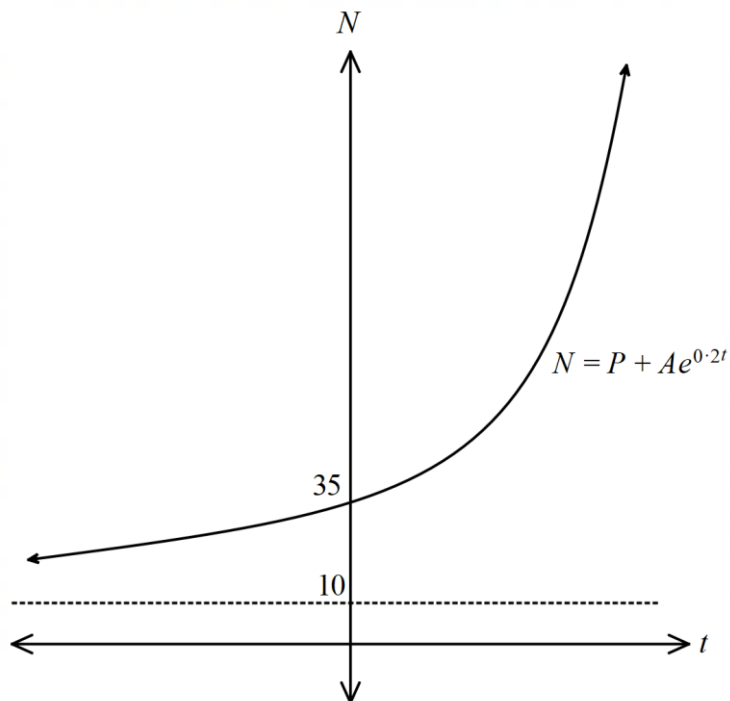
- (A) $\sqrt{10}$
- (B) $\sqrt{24}$
- (C) $\sqrt{26}$
- (D) $\sqrt{34}$

5. Consider the polynomial $p(x) = ax^3 + bx^2 + cx + d$, where $a < 0$, $b < 0$, $a \neq 0$. Which of the following curves represents $y = p(x)$?



6. Which of the following is the derivative of $\arcsin\left(\frac{x}{3}\right)$?
- (A) $\frac{1}{3\sqrt{1-x^2}}$
- (B) $\frac{1}{\sqrt{9-x^2}}$
- (C) $\frac{1}{3\sqrt{1-9x^2}}$
- (D) $-\frac{1}{3}\cos\left(\frac{x}{3}\right)$
7. How many distinct arrangements are possible from the letters of the word NECESSITIES?
- (A) 50 400
- (B) 554 400
- (C) 1 663 200
- (D) 39 916 800
8. Given the function $f(x) = x + e^x$, $x \geq 0$.
What is the gradient of the inverse function $y = f^{-1}(x)$ at $(1, 0)$?
- (A) 2
- (B) $1 + \frac{1}{e}$
- (C) $\frac{1}{2}$
- (D) $\frac{e}{e+1}$

9. The quantity N increases exponentially over time according to the equation $N = P + Ae^{0.2t}$. The graph below shows the change in N over time (t).

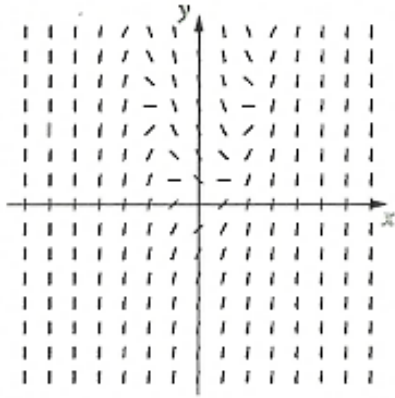


Which equation describes the rate of change of N with respect to t ?

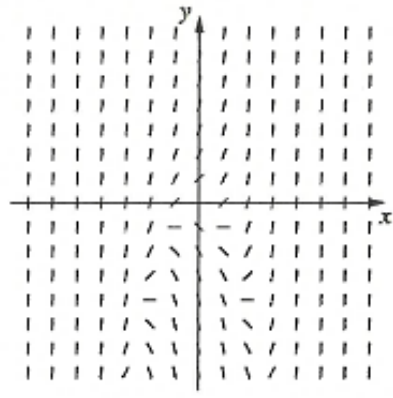
- (A) $\frac{dN}{dt} = 0.2(N - 10)$
- (B) $\frac{dN}{dt} = 0.2(N - 25)$
- (C) $\frac{dN}{dt} = 10(N - 0.2)$
- (D) $\frac{dN}{dt} = 25(N - 10)$

10. Which of the following slope fields represents the differential equation $\frac{dy}{dx} = x^2 - y$?

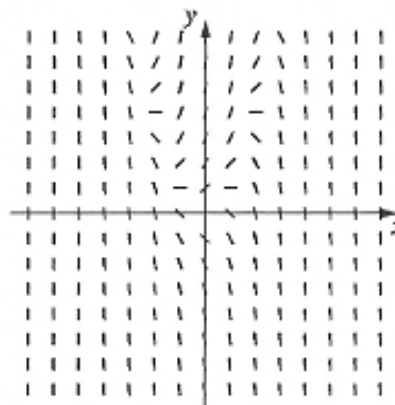
(A)



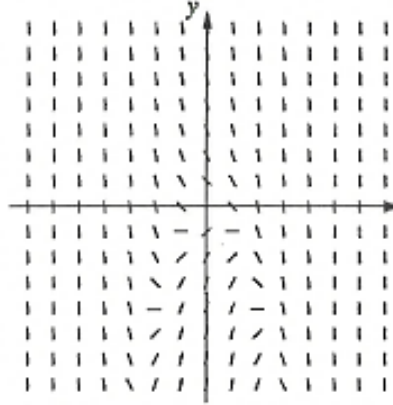
(B)



(C)



(D)



SECTION II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section.

Answer these questions in the Answer Book provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (14 marks)	Marks
(a) Given $\underline{a} = \underline{i} + 3\underline{j}$ and $\underline{b} = \underline{i} - \underline{j}$, find the vector projection of $\underline{a}$ onto $\underline{b}$.	2
(b) Solve $\frac{1}{2-x} \geq -1$.	2
(c) Using the substitution $u = 3x^2 + 1$, find $\int x \cos(3x^2 + 1) dx$.	2
(d) Find the coefficient of the term in x^6 in the expansion of $\left(\frac{3x^2}{2} + \frac{1}{5}\right)^6$.	2
(e) The sum of two of the roots of the equation $2x^3 + 6x^2 + kx - 150 = 0$ is zero. Find the value of k .	2
(f) Show that $\cot(A+B) = \frac{\cot A \cot B - 1}{\cot A + \cot B}$.	2
(g) Find the cartesian equation of the curve with the parametric equations given.	2

$$\begin{cases} x = 2 \cos \theta \\ y = \sqrt{3} \sin \theta \end{cases} \quad 0 \leq \theta \leq 2\pi$$

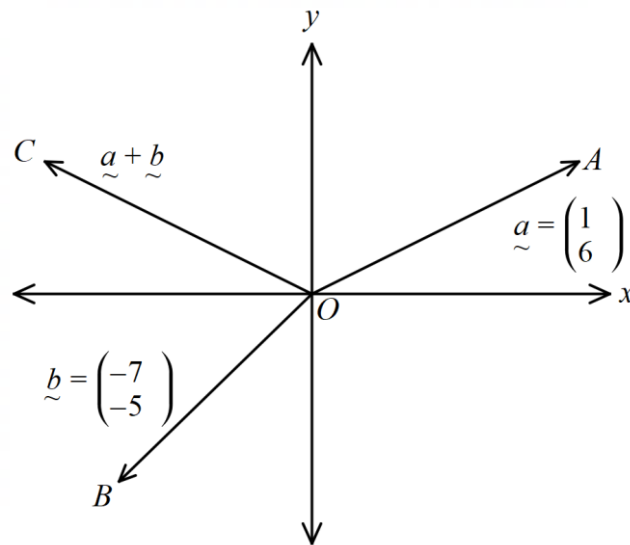
End of Question 11

Question 12 (15 marks)**Marks**

- (a) Write $\sqrt{3} \cos x - 3 \sin x$ in the format $R \cos(x + A)$ where $R > 0$ and $0 < A < \frac{\pi}{2}$.

2

- (b) The diagram shows the vectors $\underline{a} = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} -7 \\ -5 \end{pmatrix}$ and the resultant vector $\underline{a} + \underline{b}$.



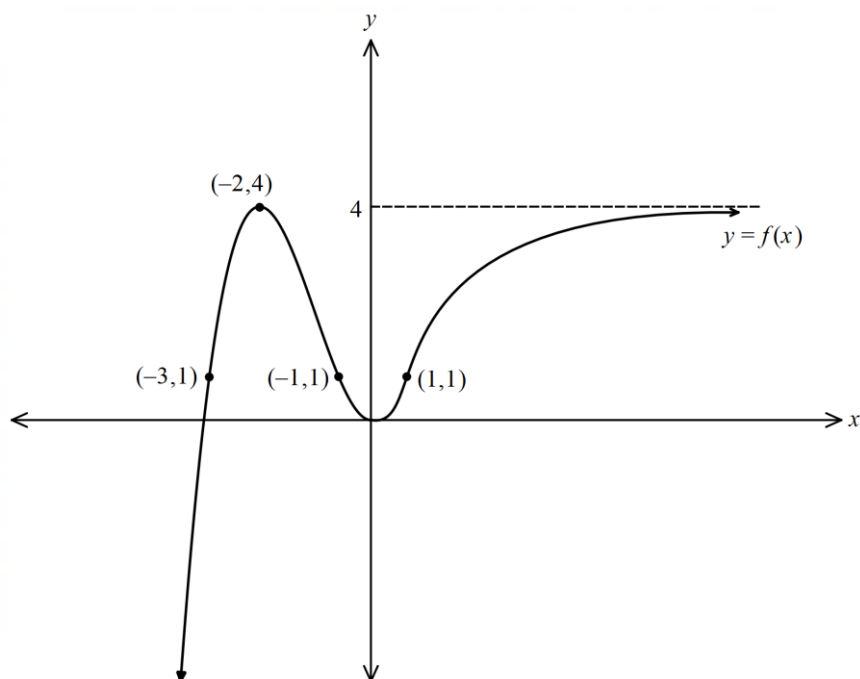
- (i) Show that the triangle AOC formed by the vectors is an isosceles triangle.
- (ii) Find the size of $\angle AOC$.

2**2****Question 12 continues on page 9**

Question 12 continued.

- (c) The diagram of $y = f(x)$ is shown.

2



On the diagram provided in the answer booklet, sketch the graph of $y = \sqrt{f(x)}$, clearly showing all turning points, intercepts, asymptotes and points of intersection with $y = f(x)$.

- (d) If the roots of the equation $x^3 + 3x^2 - 6x - 8 = 0$ are in arithmetic sequence, find all the roots.

2

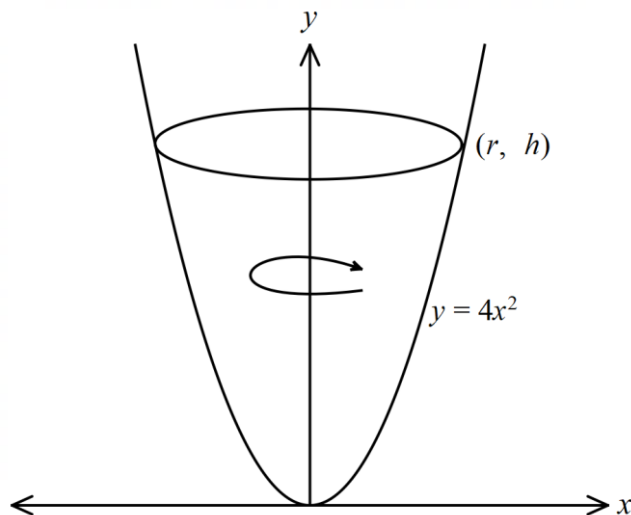
Question 12 continues on page 10

Question 12 continued.

Marks

- (e) (i) The internal surface of a water reservoir is created by rotating the parabola about the y axis between $y = 0$ and $y = h$.

2



Show that the volume of the water in the reservoir is given by $V = 2\pi r^4$.

- (ii) The variable h represents the depth of the water in the reservoir at time t hours and r metres is the radius of the upper surface of the water at that time.

1

The radius of the surface is reducing at a constant rate such that $\frac{dr}{dt} = -2$.

Show that the rate of change of the depth is given by $\frac{dh}{dt} = -16r$.

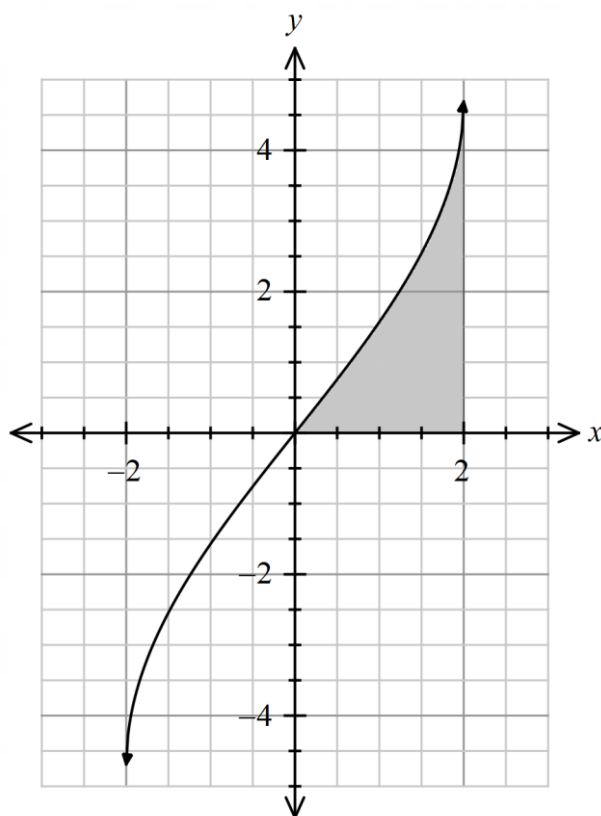
- (iii) Find the rate of change of the volume when the radius is 5 metres.

2

End of Question 12

Question 13 (16 marks)**Marks**

- (a) Evaluate $\int_0^{\frac{1}{\sqrt{3}}} \frac{\sin(\tan^{-1} x)}{1+x^2} dx$ by using the substitution $u = \tan^{-1} x$. **3**
- (b) Use mathematical induction to prove that $9^n - 4^n$ is divisible by 5 for all integers $n \geq 1$. **3**
- (c) Find the shaded area between the curve $y = 3 \sin^{-1}\left(\frac{x}{2}\right)$ and the x -axis for $x \in [0, 2]$. **3**

**Question 13 continues on page 10**

Question 13 continued.

Marks

- (d) A beehive initially had a population of 100 bees.
The population growth of the colony is given by the logistic equation

4

$$\frac{dP}{dt} = 0.2P \left(\frac{M - P}{M} \right),$$

where t is the time in months and M is the carrying capacity.

After 12 months the population of the hive was 400 bees.

Use the fact that $\frac{M}{P(M - P)} = \frac{1}{P} + \frac{1}{M - P}$ to show that the carrying capacity, M , is approximately 570 bees.

- (e) It is known that $\cos 3\theta = 4 \cos \theta \cos \left(\frac{\pi}{3} - \theta \right) \cos \left(\frac{\pi}{3} + \theta \right)$.

3

Using this result, or otherwise, solve $\sqrt{2} \cos \theta (\cos^2 \theta - 3 \sin^2 \theta) = 1$ for $0 \leq \theta \leq 2\pi$.

End of Question 13

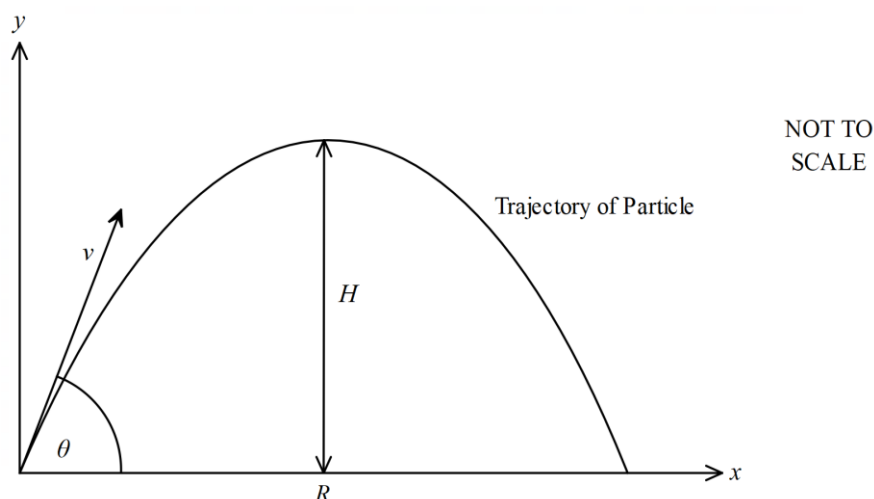
Question 14 (15 marks)**Marks**

- (a) Given that $\frac{dy}{dx} = \frac{y-1}{x+2}$ and $y(6) = 3$, find the value of $y(1)$. **3**
- (b) Two circular tables seat 6 and 4 people respectively. In how many ways can 10 people be seated at the two tables if Taylor and Travis are to sit together at the bigger table? **2**
- (c) (i) Solve for $\tan 4\alpha = 1$ for $0 \leq \alpha \leq \pi$. **1**
- (ii) Given $\tan 4\alpha = \frac{4 \tan \alpha - 4 \tan^3 \alpha}{1 - 6 \tan^2 \alpha + \tan^4 \alpha}$ and by letting $x = \tan \alpha$, **2**
solve the equation $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$.
- (iii) Hence, show that $\tan^2 \frac{\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{9\pi}{16} + \tan^2 \frac{13\pi}{16} = 28$. **2**

Question 14 continues on page 12

- (d) The position vector of a particle projected from the origin at an initial velocity of v m/s at an angle of θ° to the horizontal is given by:

$$r(t) = \begin{pmatrix} vt \cos \theta \\ vt \sin \theta - \frac{gt^2}{2} \end{pmatrix}$$

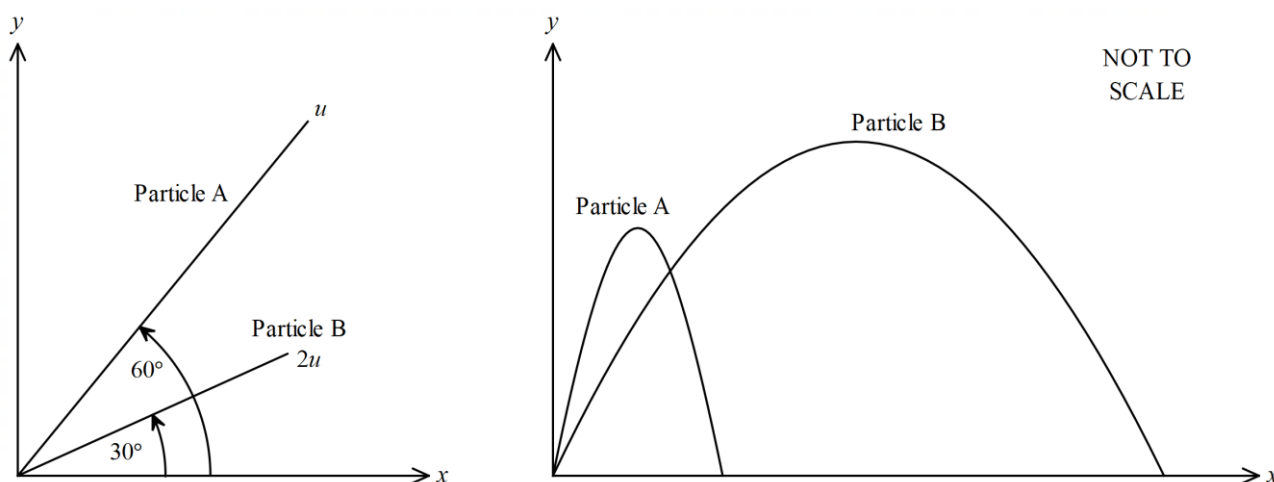


- (i) Show that the time of flight of the particle is given by $t = \frac{2v \sin \theta}{g}$ and the horizontal range by $R = \frac{v^2 \sin 2\theta}{g}$.

3

- (ii) Particle A is projected from the origin at a velocity of u m/s at an angle of 60° to the horizontal. For particle B, the initial velocity is doubled to $2u$ m/s and the angle of elevation is halved to 30° .

2



Show that the range of Particle B is 4 times that of Particle A.

END OF PAPER

Name:.....

Student
Number:

Teacher's Name:.....



Pymble Ladies' College

Mathematics Extension 1

HSC Trial Examination

Term 3 2025

ANSWER BOOKLET

- **Put your name, Student Number and Teacher's name on this Booklet.**
- For Questions 1-10 use the Multiple-Choice Answer Sheet provided.
- Answer Questions 11-14 in the spaces allocated for each question as these spaces provide guidance for the expected length of your response.
- Write using a non-erasable black or blue pen. Black is preferred.
- Extra writing space is provided at the end of each question.
- Calculators approved by NESA may be used.

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Mathematics Extension 1

HSC Trial Examination

2025

Name:.....

Student Number:

Teacher's Name:.....

Section I – Multiple Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☐ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒
B ☒
C ☐
D ☐

correct
↓

- | | | | | |
|----|------------------------------------|------------------------------------|------------------------------------|-------------------------|
| 1 | A ☒ | B ☐ | C ☐ | D ☐ |
| 2 | A ☐ | B ☒ | C ☐ | D ☐ |
| 3 | A ☐ | B ☒ | C ☐ | D ☐ |
| 4 | A ☒ | B ☐ | C ☐ | D ☐ |
| 5 | A ☐ | B ☐ | C ☒ | D ☐ |
| 6 | A ☐ | B ☒ | C ☐ | D ☐ |
| 7 | A ☐ | B ☒ | C ☐ | D ☐ |
| 8 | A ☐ | B ☐ | C ☒ | D ☐ |
| 9 | A ☒ | B ☐ | C ☐ | D ☐ |
| 10 | A ☒ | B ☐ | C ☐ | D ☐ |

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SECTION II

60 marks

Your responses should include relevant mathematical reasoning and/or calculations.
Extra writing space is provided at the end of each question.

Question 11 (14 marks)

Marks

(a)

$$\text{Proj}_k a = \frac{a \cdot k}{|k|^2} k$$

2

$$= \frac{1-3}{(\sqrt{2})^2} (1-j)$$

1 - progress

$$= -(1-j)$$

2 - correct solutions.

$$= -1 + j$$

(b)

$$\frac{1}{2-x} \geq -1 \quad (x \neq 2).$$

2

$$2-x \geq -(2-x)^2.$$

1 - get to *

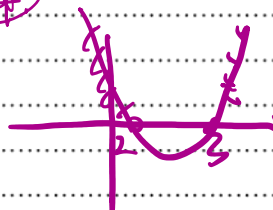
$$(2-x) + (2-x)^2 \geq 0.$$

2 - correct solutions

$$(2-x)(1+2-x) \geq 0.$$

$$(2-x)(3-x) \geq 0. \quad (*)$$

$$(x-3)(x-2) \geq 0.$$



$$\therefore x < 2 \text{ or } x \geq 3$$

Question 11 continues on page 2

Question 11 continued.

Marks

(c)

2

$$\begin{aligned} & \int x \cos(3x^2+1) dx \\ &= \int \cos(u) \cdot \frac{du}{6} \\ &= \frac{1}{6} (\sin u) + C. \\ &= \frac{1}{6} \sin(3x^2+1) + C. \end{aligned}$$

1 - progress

2 - correct solutions

(d)

2

$$\begin{aligned} & \left(\frac{2x^2}{2} + \frac{1}{5} \right)^6 \\ &= \sum_{r=0}^6 {}^6C_r \left(\frac{2}{2} x^2 \right)^{6-r} \left(\frac{1}{5} \right)^r \\ &= \sum_{r=0}^6 {}^6C_r \left(\frac{2}{2} \right)^{6-r} \left(\frac{1}{5} \right)^r x^{12-2r} \end{aligned}$$

let $x^{12-2r} = x^6$

$12-2r = 6$

$r = 3$

- Coef of $x^6 = {}^6C_3 \left(\frac{2}{2} \right)^3 \left(\frac{1}{5} \right)^3$

$= \frac{27}{125} \quad (= 0.54)$

1 - expansion of binomial

2 - correct solutions

Question 11 continues on page 3

Question 11 continued.

Marks

- (e) Let the roots be α , $-\alpha$ and β of $2x^3 + 6x^2 + kx - 150 = 0$.
Sum of roots $= \alpha + (-\alpha) + \beta = -\frac{6}{2}$
 $\beta = -3$
 $2(-3)^3 + 6(-3)^2 - 3k - 150 = 0$
 $-3k = 150$
 $k = -50$

2	Finding k correctly
1	Finding one root

Question 11 continues on page 4

Marks

(f)

2	fully correct
1	Correct process of breaking up LHS OR RHS

4

Question 11 continued.

Marks

(g)

2

$$\begin{cases} x = 2 \cos \theta \Rightarrow \cos \theta = \frac{x}{2} \\ y = \sqrt{3} \sin \theta \Rightarrow \sin \theta = \frac{y}{\sqrt{3}} \end{cases}$$

$$\text{Since } \sin^2 \theta + \cos^2 \theta = 1,$$

$$\text{so } \left(\frac{x}{2}\right)^2 + \left(\frac{y}{\sqrt{3}}\right)^2 = 1$$

$$\frac{x^2}{4} + \frac{y^2}{3} = 1$$

2	Simplifying to answer
1	Correctly using $\sin^2 \theta + \cos^2 \theta = 1$ in terms of x and y

End of Question 11

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Question 12 (15 marks)

Marks

(a)

2

$$\begin{aligned}\sqrt{3} \cos x - 3 \sin x &= R \cos(x + A) \\ R^2 &= (\sqrt{3})^2 + 3^2 & \tan A &= \frac{3}{\sqrt{3}} = \sqrt{3} \\ R &= \sqrt{12} & A &= \frac{\pi}{3} \\ &= 2\sqrt{3}\end{aligned}$$

$$\therefore \sqrt{3} \cos x - 3 \sin x = 2\sqrt{3} \cos\left(x + \frac{\pi}{3}\right)$$

2	correct answer
1	getting R or A correctly

Question 12 continues on page 10

(b)

$$(i) \quad \vec{OC} = \vec{a} + \vec{b} = \vec{c} \quad \left| \quad \vec{OA} = \vec{a} \right. \quad 2$$

$$= \begin{bmatrix} -6 \\ 1 \end{bmatrix} \quad \left| \quad = \begin{bmatrix} 1 \\ 6 \end{bmatrix} \right.$$

$$|\vec{OC}| = \sqrt{(-6)^2 + 1^2} \quad \left| \quad |\vec{OA}| = \sqrt{1^2 + 6^2} \right.$$

$$= \sqrt{37} \quad \left| \quad = \sqrt{37} \right.$$

$$\text{Since } |\vec{OC}| = |\vec{OA}|,$$

$\therefore \triangle AOC$ is an isosceles triangle.

2	getting $ \vec{OC} = \vec{OA} $
1	found $\vec{OC}$ and $ \vec{OA} $ correctly

$$(ii) \quad \cos \angle AOC = \frac{\vec{a} \cdot \vec{c}}{|\vec{a}| |\vec{c}|} \quad 2$$

$$= \frac{(1)(-6) + (6)(1)}{\sqrt{37} \times \sqrt{37}}$$

$$= \frac{0}{37}$$

$$= 0$$

$$\text{Since } \cos \angle AOC = 0,$$

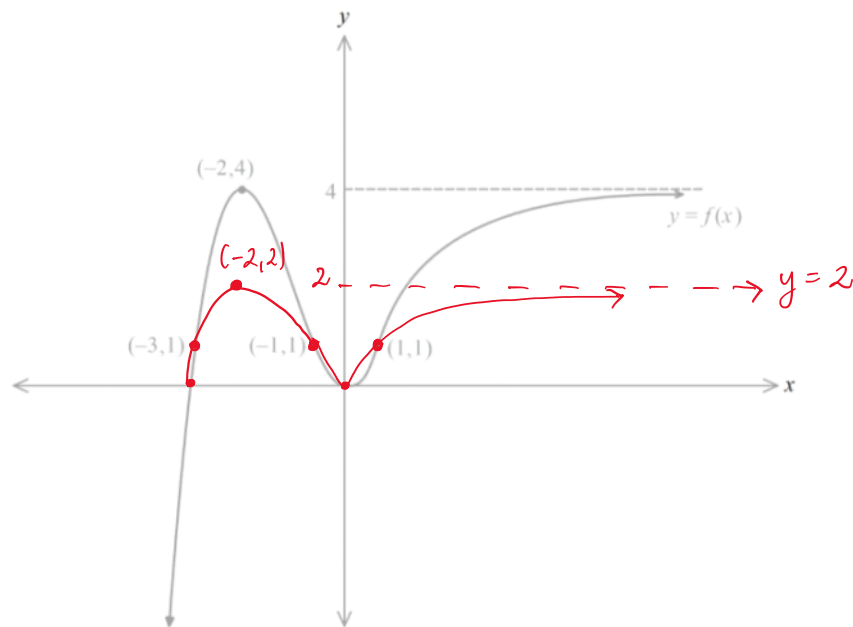
$$\angle AOC = \frac{\pi}{2}.$$

2	getting the angle correctly
1	getting $\vec{a} \cdot \vec{c} = 0$

Question 12 continues on page 11

(c)

2



- [2] marks - everything correct. Accept only the error where $(-2, 4)$ misaligned with $y = 2$ asymptote
- [1] mark - everything correct but extended graph further to left incorrect
- correct shape but fail to pass through $y = 1$.
 - correct shape but fail to note asymptote $y = 2$
 - correct shape but fail to note $(-2, 2)$.
- [0] mark - incorrect shape
- made more than 2 errors
 - only drew one portion of graph.

Question 12 continues on page 12

(d) Let roots be $\alpha-d, \alpha, \alpha+d$

2

$$\alpha-d + \alpha + \alpha+d = -3$$

$$3\alpha = -3$$

$$\alpha = -1$$

*

$$\alpha(\alpha-d)(\alpha+d) = 8$$

$$(1-d^2) = 8$$

$$d^2 = 9$$

$$d = \pm 3$$

$\therefore$ roots are $-4, -1, 2$

[1] mark get to

*

 or equivalent working out.

[2] marks get

 with correct working out.

Question 12 continues on page 13

Question 12 continued.

Marks

(e) (i)

2

$$V = \pi \int_0^h \frac{y}{4} dy$$

$$= \frac{\pi}{4} \left[\frac{y^2}{2} \right]_0^h$$

$$= \frac{\pi}{8} [h^2 - 0]$$

$$= \frac{\pi h^2}{8}$$

*

$$\text{But } h = 4r^2$$

$$V = \frac{\pi (4r^2)^2}{8}$$

**

$$\therefore V = 2\pi r^4$$

1 mark *

2 marks ** with correct working out.

Question 12 continues on page 14

Question 12 (part e) continued.

Marks

(ii) $h = 4r^2$ 1

$$\frac{dh}{dr} = 8r$$

$$\frac{dh}{dt} = \frac{dh}{dr} \times \frac{dr}{dt}$$

$$= 8r \times -2$$

$$= -16r$$

(iii) $V = 2\pi r^4$ 2

$$\frac{dV}{dr} = 8\pi r^3$$

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$= 8\pi r^3 \times -2$$

$$= -16\pi r^3$$

When $r = 5$

$$\frac{dV}{dt} = -16\pi (5)^3$$

$$= -2000\pi$$

✖ ✖ 2 marks

✖ 1 mark

End of Question 12

[illegible]

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Question 13 (16 marks)

Marks

(a)

3

$$\int_0^{\frac{1}{\sqrt{3}}} \frac{\sin(\tan^{-1} x)}{1+x^2} dx$$

$$= \int_0^{\frac{\pi}{6}} (\sin u) du$$

$$= [-\cos u]_0^{\frac{\pi}{6}}$$

$$= -\cos \frac{\pi}{6} + \cos 0$$

$$= -\frac{\sqrt{3}}{2} + 1$$

$$u = \tan^{-1} x$$

$$du = \frac{1}{1+x^2} dx$$

$$\text{When } x = \frac{1}{\sqrt{3}}$$

$$u = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

$$= \frac{\pi}{6}$$

$$\text{When } x = 0,$$

$$u = \tan^{-1} 0 = 0$$

3 | fully correct

2 | putting * together as an integral with one variable

1 | finding * correctly

Question 13 continues on page 18

(b)

Let the statement be $(9^n - 4^n)$ is divisible by 5,
for $n \geq 1, n \in \mathbb{Z}$.

$$\begin{aligned} \text{For } n=1, \quad 9^n - 4^n \\ &= 9 - 4 \\ &= 5 \quad \text{which is divisible by 5.} \end{aligned}$$

$\therefore$ The statement is true for $n=1$.

Assume the statement is true for $n=k, k \geq 1$.

$$\text{i.e. } 9^k - 4^k = 5M, \quad M \in \mathbb{Z}.$$

To prove it's true for $n=k+1$.

$$\text{i.e. } 9^{k+1} - 4^{k+1} = 5N, \quad N \in \mathbb{Z}.$$

$$\begin{aligned} \text{LHS} &= 9^{k+1} - 4^{k+1} \\ &= 9 \cdot 9^k - 4 \cdot 4^k \\ &= 9 \cdot 9^k - 9 \cdot 4^k + 5 \cdot 4^k \\ &= 9(9^k - 4^k) + 5 \cdot 4^k \\ &= 9(5M) + 5 \cdot 4^k \quad (\text{from assumption}) \\ &= 5(9M + 4^k). \end{aligned}$$

$$\text{Since } M, k \in \mathbb{Z} \therefore (9M + 4^k) \in \mathbb{Z}.$$

$$\begin{aligned} \therefore \text{LHS} &= 5(9M + 4^k) \\ &= 5N \quad \text{where } N = 9M + 4^k, N \in \mathbb{Z}. \end{aligned}$$

$\therefore$ The statement is true for $n=k+1$ if it's true for $n=k$.

Since the statement is true for $n=1$, it's true for $n=1+1=2, n=2+1=3$ & so on. By Mathematical induction, it's true for all $n \geq 1, n \in \mathbb{Z}$.

1 — Correct proof for $n=1$.

2 — Good set up for the assumption, conclusion, working towards the proof.

3 — Correct solutions.

Question 13 continues on page 19

(c)

3

$$y = 3\sin^{-1}\left(\frac{x}{2}\right) \quad \text{At } x = -2, y = -\frac{3\pi}{2}$$

$$\text{At } x = 2, y = \frac{3\pi}{2}$$

$$\frac{y}{3} = \sin^{-1}\left(\frac{x}{2}\right)$$

$$\frac{x}{2} = \sin\left(\frac{y}{3}\right)$$

$$x = 2\sin\left(\frac{y}{3}\right)$$

$$\left(-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}\right)$$

$$\text{Area} = 2 \times \left(\frac{3\pi}{2}\right) - \int_0^{\frac{3\pi}{2}} 2\sin\left(\frac{y}{3}\right) dy$$

$$= 3\pi - (2 \times 3) \left[-\cos\left(\frac{y}{3}\right)\right]_0^{\frac{3\pi}{2}}$$

$$= 3\pi + 6 \left(\cos\frac{\pi}{2} - \cos(0)\right)$$

$$= (3\pi - 6) \text{ unit}^2$$

$$1 - \text{Convert to } x = 2\sin\left(\frac{y}{3}\right)$$

$$2 - \text{Progress to areas}$$

$$3 - \text{correct solutions.}$$

Question 13 continues on page 20

Question 13 continued.

(d)

4

$$\begin{aligned} \frac{dp}{dt} &= 0.2P \left(\frac{m-p}{m} \right) \\ \int \frac{m}{P(m-P)} dp &= \int 0.2 dt \quad (A) \\ \int \frac{1}{P} + \frac{1}{m-P} dp &= \int 0.2 dt \\ \ln|P| - \ln|m-P| &= 0.2t + C \\ \ln \left| \frac{P}{m-P} \right| &= 0.2t + C \\ e^{0.2t+C} &= \left| \frac{P}{m-P} \right| \\ e^{0.2t} \cdot e^C &= \left| \frac{P}{m-P} \right| \\ \text{let } e^C = A, A \text{ pos or neg} \\ A e^{0.2t} &= \left| \frac{P}{m-P} \right| \quad (1) \end{aligned}$$

$$\begin{aligned} t=0, P=100 \\ A e^0 &= \left| \frac{100}{m-100} \right| \\ A &= \frac{100}{m-100}, A > 0 \quad (2) \end{aligned}$$

$$(1) \quad A e^{0.2t} = \frac{P}{m-P}, A > 0$$

$$\begin{aligned} t=12, P=400 \\ A e^{2.4} &= \frac{400}{m-400} \end{aligned}$$

$$\begin{aligned} \frac{100}{m-100} e^{2.4} &= \frac{400}{m-400} \\ \frac{e^{2.4}}{m-100} &= \frac{4}{m-400} \\ m e^{2.4} - 400 e^{2.4} &= 4m - 400 \\ m(e^{2.4} - 4) &= 400 e^{2.4} - 400 \end{aligned}$$

$$\begin{aligned} m &= \frac{400 e^{2.4} - 400}{e^{2.4} - 4} \\ &= 570.862.. \\ &\approx 570 \text{ bees} \end{aligned}$$

Question 13 continues on page 21

Question 13 part (d) continued.

Marks[illegible]

Question 13 continues on page 22

Question 13 continued.

Marks

3

(e)

$$\cos\left(\frac{\pi}{3}-\theta\right) = \cos\frac{\pi}{3} \cdot \cos\theta + \sin\frac{\pi}{3} \cdot \sin\theta$$

$$= \frac{1}{2} \cos\theta + \frac{\sqrt{3}}{2} \sin\theta$$

$$\cos\left(\frac{\pi}{3}+\theta\right) = \cos\frac{\pi}{3} \cdot \cos\theta - \sin\frac{\pi}{3} \cdot \sin\theta$$

$$= \frac{1}{2} \cos\theta - \frac{\sqrt{3}}{2} \sin\theta$$

$$\text{so } \cos\left(\frac{\pi}{3}-\theta\right) \cdot \cos\left(\frac{\pi}{3}+\theta\right)$$

$$= \frac{1}{4} \cos^2\theta - \frac{3}{4} \sin^2\theta$$

$$\text{so } 4 \cos\left(\frac{\pi}{3}-\theta\right) \cdot \cos\left(\frac{\pi}{3}+\theta\right)$$

$$= (\cos^2\theta - 3 \sin^2\theta)$$

$$\text{so } 4 \cos\theta \cdot \cos\left(\frac{\pi}{3}-\theta\right) \cdot \cos\left(\frac{\pi}{3}+\theta\right)$$

$$= \cos\theta (\cos^2\theta - 3 \sin^2\theta)$$

$$\text{so } \boxed{1} \cos\theta (\cos^2\theta - 3 \sin^2\theta) = \cos 3\theta$$

$$\boxed{2} \rightarrow \cos\theta (\cos^2\theta - 3 \sin^2\theta) = \frac{1}{\sqrt{2}}$$

$$\text{so } \cos 3\theta = \frac{1}{\sqrt{2}} \quad \times$$

$$3\theta = \frac{\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}, \frac{15\pi}{4}, \frac{17\pi}{4}, \frac{23\pi}{4}$$

$$\theta = \frac{\pi}{12}, \frac{7\pi}{12}, \frac{9\pi}{12}, \frac{15\pi}{12}, \frac{17\pi}{12}, \frac{23\pi}{12}$$

----- also -----

$$\cos 3\theta = 4 \cos\theta \cdot \left[\cos\left(\frac{\pi}{3}-\theta\right) \cdot \cos\left(\frac{\pi}{3}+\theta\right) \right]$$

$$= 4 \cos\theta \cdot \frac{1}{2} [\cos(2\theta) + \cos\left(\frac{2\pi}{3}\right)]$$

$$= 2 \cos\theta \cdot (2 \cos^2\theta - 1 - \frac{1}{2})$$

$$= 2 \cos\theta \cdot (2 \cos^2\theta - \frac{3}{2} (\cos^2\theta + \sin^2\theta))$$

$$= 2 \cos\theta \cdot (\frac{1}{2} \cos^2\theta - \frac{3}{2} \sin^2\theta)$$

$$\cos 3\theta = \cos\theta (\cos^2\theta - 3 \sin^2\theta)$$

$$\text{so } \sqrt{2} \cos 3\theta = 1$$

$\boxed{1}$ expando
 $\cos\left(\frac{\pi}{3}-\theta\right)$ or
 $\cos\left(\frac{\pi}{3}+\theta\right)$
 • attempts other
 identities to solve

$$\boxed{2} \cos\left(\frac{\pi}{3}-\theta\right) \cdot \cos\left(\frac{\pi}{3}+\theta\right)$$

$$= \frac{1}{4} \cos^2\theta - \frac{3}{4} \sin^2\theta$$

$\boxed{3} \cos 3\theta = \frac{1}{\sqrt{2}}$
 and
 $\theta = \frac{\pi}{12} \dots \text{etc.}$

End of Question 13

This image shows a full page of a document template designed for handwriting practice or general note-taking. It consists of approximately 28 evenly spaced horizontal dotted lines across the entire width of the page. The background is plain white, and there are no margins, headers, footers, or other markings present.

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Question 14 (15 marks)

Marks

(a)

3

$$\begin{aligned}\frac{dy}{dx} &= \frac{y-1}{x+2} \\ \int \frac{dy}{y-1} &= \int \frac{dx}{x+2} \\ \ln|y-1| &= \ln|x+2| + c \\ y(6) &= 3 \\ \ln 2 &= \ln 8 + c \\ c &= \ln \frac{1}{4} \\ \ln|y-1| - \ln|x+2| &= \ln \frac{1}{4} \\ \therefore \ln \left| \frac{y-1}{x+2} \right| &= \ln \frac{1}{4}\end{aligned}$$

$$\left| \frac{y-1}{x+2} \right| = \frac{1}{4}$$

for $y(1)$

$$\therefore \left| \frac{y-1}{3} \right| = \frac{1}{4}$$

$$|y-1| = \frac{3}{4}$$

$$y-1 = -\frac{3}{4} \text{ or } y-1 = \frac{3}{4}$$

$$y = \frac{1}{4} \text{ or } y = \frac{7}{4}$$

$$\begin{aligned}\int_3^y \frac{1}{y-1} dy &= \int_6^1 \frac{1}{x+2} dx \\ [\ln|y-1|]_3^y &= [\ln|x+2|]_6^1 \\ \ln|y-1| - \ln 2 &= \ln 3 - \ln 8 \\ \ln|y-1| &= \ln 3 - \ln 8 + \ln 2 \\ \ln|y-1| &= \ln \frac{3}{4}\end{aligned}$$

$$\boxed{1} \quad \ln|y-1| = \ln|x+2| + c$$

$$\boxed{2} \quad \left| \frac{y-1}{x+2} \right| = \frac{1}{4}$$

attempts to use $y(1)$
to solve
 $|y-1| = \frac{3}{4}$

$$\boxed{3} \quad y = \frac{1}{4} \text{ or } y = \frac{7}{4}$$

or

$$\boxed{1} \quad [\ln|y-1|]_3^y = [\ln|x+2|]_6^1$$

$$\boxed{2} \quad \ln|y-1| = \ln \frac{3}{4}$$

attempts to solve
 $|y-1| = \frac{3}{4}$

$$\boxed{3} \quad y = \frac{1}{4} \text{ or } y = \frac{7}{4}$$

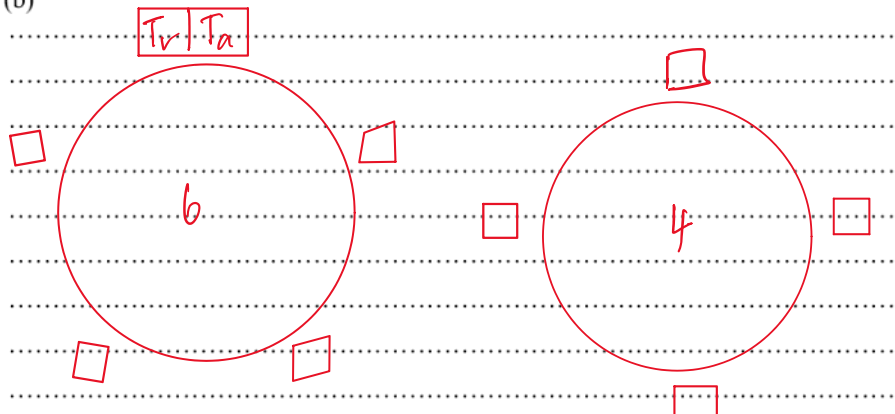
Question 14 continues on page 26

Question 14 continued.

Marks

(b)

2



$$= {}^8C_4 \times 2! \times 4! \times 3!$$

$$= 3360 \times 6$$

$$= 20160$$

[0] mark — more than 3 errors.

Question 14 continues on page 27

[1] mark — forgot to treat T_r and T_a seated together $\therefore 50400$

— forgot to include $2!$

— one mistake eg. ${}^8C_4 \times 2! \times 4! \times 4! = 80640$

[2] marks — correct answer

Question 14 continued.

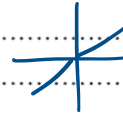
Marks

(c) (i)

$$\tan 4\alpha = 1 \quad 0 \leq \alpha \leq \pi$$

1

$$0 \leq 4\alpha \leq 4\pi$$



$$4\alpha = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$$

$$\alpha = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16}$$

1 | correct solution

(ii)

$$x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$$

2

$$1 - 6x^2 + x^4 = 4x - 4x^3$$

$$1 = \frac{4x - 4x^3}{1 - 6x^2 + x^4}$$

$$\text{for } x = \tan \alpha$$

$$1 = \tan 4\alpha$$

$$\text{from (i)} \quad \alpha = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16}$$

$$\therefore x = \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, \tan \frac{9\pi}{16}, \tan \frac{13\pi}{16}$$

Question 14 continues on page 28

1 | uses identity to relate to (i), $\alpha =$
2 | uses substitution $x = \tan \alpha$ to get sol.

(iii)

sums/products of roots

2

$$x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$$

$$\begin{aligned} \alpha^2 + \beta^2 + \gamma^2 + \delta^2 &= (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta) \\ &= (\text{sum of roots})^2 - 2(\text{sum of roots two at a time}) \\ &= \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right) \\ &= (-4)^2 - 2(-6) \\ &= 16 + 12 \\ &= 28 \end{aligned}$$

$$\therefore \tan^2 \frac{\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{9\pi}{16} + \tan^2 \frac{13\pi}{16} = 28$$

- | | |
|---|---|
| 1 | correct $\left(-\frac{b}{a}\right)$ or $\left(\frac{c}{a}\right)$ (identifying sums of roots) |
| 2 | correctly showing |

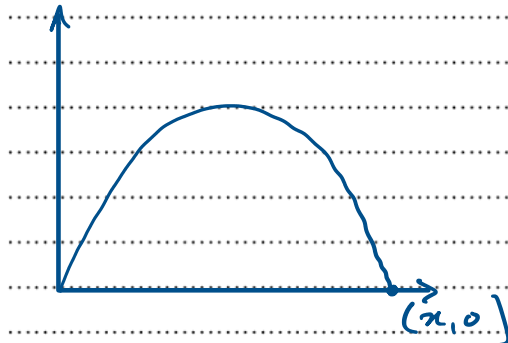
Note: Students should be identifying a sum/products of roots question and automatically finding $-\frac{b}{a}$, $\frac{c}{a}$ etc as required.

Question 14 continues on page 29

(d)

$$(i) \quad r(t) = Vt \cos \theta \, \underline{i} + \left(Vt \sin \theta - \frac{gt^2}{2} \right) \underline{j}$$

3



Flight: $0 = Vt \sin \theta - \frac{gt^2}{2}$ Range: $x = Vt \cos \theta$
 when $t = \frac{2V \sin \theta}{g}$
 $0 = 2Vt \sin \theta - gt^2$
 $0 = t(2V \sin \theta - gt)$
 $t = 0$ (initial time) $\therefore t = \frac{2V \sin \theta}{g}$
 $x = V \left[\frac{2V \sin \theta}{g} \right] \cos \theta$
 $= \frac{V^2 \cdot 2 \sin \theta \cos \theta}{g}$

1 | one correct
 2 | both correct.

$$\therefore x = \frac{V^2 \sin 2\theta}{g}$$

note: For a quadratic equation (in t)
 you should be dividing by the
 variable as you lose a solution.
 A justification for not taking $t=0$
 should be provided.

Question 14 continues on page 30

(ii) A: $v = u$ $\theta = 60^\circ$

B: $v = 2u$ $\theta = 30^\circ$

2

$$x_A = \frac{u^2 \sin[2 \cdot 60^\circ]}{g}$$

$$x_B = \frac{(2u)^2 \sin[2 \cdot 30^\circ]}{g}$$

$$= \frac{u^2 \sin 120^\circ}{g}$$

$$= \frac{4u^2 \sin 60^\circ}{g}$$

$$= \frac{u^2 \frac{\sqrt{3}}{2}}{g}$$

$$= \frac{4u^2 \frac{\sqrt{3}}{2}}{g}$$

$$= \frac{u^2 \sqrt{3}}{2g}$$

$$= \frac{2u^2 \sqrt{3}}{g}$$

$$= 4 \cdot \frac{u^2 \sqrt{3}}{2g}$$

$$1 \quad \frac{u^2 \sqrt{3}}{2g} \text{ or } \frac{2u^2 \sqrt{3}}{g}$$

$$= 4x_A$$

2 correct

 $\therefore$ 4 times the range of A.

NOTE: Many students not using the expression for the range from part (i), and doing a lot of work to derive it again with the provided information. More attention needs to be paid to linking subsequent parts of questions.

END OF PAPER